PUSHOUTS ALONG A CLOSED IMMERSION AND AN INTEGRAL MORPHISM ARE COMPATIBLE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19901-19943, immediately following the statement at 19875-19899.

Editorial note. Five cross-chapter pointers (Limits, More on Algebra, Morphisms) plus two same-chapter statements are recorded in assembly.external_references.

Editorial note. The separatedness argument is displayed in full, including the factorisation used to inherit separatedness.

Allowed standard prerequisites (editorial).

- The situation of a pushout along a closed immersion and an integral morphism. chapter situation situation-pushout-along-closed-immersion-and-integral, cited in the statement. Fixes the setup whose base change behaviour is being asserted.
- Existence and properties of the pushout in the situation. chapter proposition proposition-pushout-along-closed-immersion-and-integral, cited at proof line 19938. Supplies existence of the pushout and the fact that Y' coproduct X' -> Y' coproduct_{Z'} X' is universally closed and surjective.
- Separatedness descends along a universally closed surjection. Morphisms, lemma morphisms-lemma-image-universally-closed-separated, cited at proof lines 19941-19942. Concludes separatedness of the pushed-out morphism from separatedness of Y' coproduct X' -> Y coproduct_Z X.
- Local quasi-finiteness at a point. Morphisms, lemma morphisms-lemma-quasi-finite-at-point-characterize, cited at proof lines 19930-19931. Certifies local quasi-finiteness of the base changed pushout morphism.
- Modules and algebras over a fibre product. More on Algebra, lemmas more-algebra-lemma-module-over-fibre-product and more-algebra-lemma-properties-algebras-over-fibre-product, cited in the proof. Describe the pushout algebra after base change, which is the algebraic content of the compatibility.

Source reference. Original lemma 0AP9: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19681–19681. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0E21: focused-seeds/geometry/S9-morph-pushout-functor/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07RU: focused-seeds/geometry/S9-morph-pushout-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof

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cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08KQ: focused-seeds/geometry/S9-morph-pushout-functor/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09MQ: focused-seeds/geometry/S9-morph-pushout-functor/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TH: focused-seeds/geometry/S9-morph-pushout-functor/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 0E25: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19758–19758. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19708. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0ECI suppose given a commutative diagram*

$$\begin{array}{ccccc} Y' & \xleftarrow{j'} & Z' & \xrightarrow{i'} & X' \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle f} \\ Y & \longleftarrow & Z & \longrightarrow & X \end{array}$$

with cartesian squares and f, g, h separated and locally quasi-finite. Then

(1) *the pushouts $Y \amalg_Z X$ and $Y' \amalg_{Z'} X'$ exist,*
(2) *$Y' \amalg_{Z'} X' \to Y \amalg_Z X$ is separated and locally quasi-finite, and*
(3) *the squares*

$$\begin{array}{ccccc} Y' & \longrightarrow & Y' \amalg_{Z'} X' & \longleftarrow & X' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longrightarrow & Y \amalg_Z X & \longleftarrow & X \end{array}$$

are cartesian.

Proof. The pushout $Y \amalg_Z X$ exists by Proposition 0E25. To see that the pushout $Y' \amalg_{Z'} X'$ exists, we check condition (3) of Situation 0ECI holds for (X', Y', Z', i', j'). Namely, let $y' \in Y'$ and denote $y \in Y$ the image. Choose $U \subset X$ affine open with $i(j^{-1}(y)) \subset U$. Choose a quasi-compact open $U' \subset X'$ contained in $f^{-1}(U)$ containing the quasi-compact subset $i'((j')^{-1}(\{y'\}))$. By Lemma 0AP9 we see that U' is quasi-affine. Since $Z'_{y'}$ is the spectrum of an algebra integral over a field, we can

apply Limits, Lemma 0E21 and we find there exists an affine open subscheme of U' containing $i'((j')^{-1}(\{y'\}))$ as desired.

Having verified existence we check the other assertions. Affine locally we are exactly in the situation of More on Algebra, Lemma 08KQ with $B \to D$ and $A' \to C'$ locally quasi-finite[1]. In particular, the morphism $Y' \amalg_{Z'} X' \to Y \amalg_Z X$ is locally of finite type. The squares in of the diagram are cartesian by More on Algebra, Lemma 07RU. Since being locally quasi-finite can be checked on fibres (Morphisms, Lemma 01TH) we conclude that $Y' \amalg_{Z'} X' \to Y \amalg_Z X$ is locally quasi-finite.

We still have to check $Y' \amalg_{Z'} X' \to Y \amalg_Z X$ is separated. Observe that $Y' \amalg X' \to Y' \amalg_{Z'} X'$ is universally closed and surjective by Proposition 0E25. Since also the morphism $Y' \amalg X' \to Y \amalg_Z X$ is separated (as it factors as $Y' \amalg X' \to Y \amalg X \to Y \amalg_Z X$) we conclude by Morphisms, Lemma 09MQ. □

[1]To be precise $X, Y, Z, Y \amalg_Z X, X', Y', Z', Y' \amalg_{Z'} X'$ correspond to $A', B, A, B', C', D, C, D'$.